



Solve quadratic equations using the quadratic formula

- 1 The standard form of a quadratic equation is $ax^2 + bx + c = 0$.
Identify the values of a , b , and c in the quadratic equation $-8x^2 + 3x = 0$.
- 2 Using the standard form of a quadratic equation with $a > 0$, identify the values of a , b , and c in the following quadratic equations.
 - a $x^2 + 7x + 10 = 0$
 - b $x^2 - 3x - 4 = 0$
 - c $2x^2 + 9x = 0$
 - d $8x^2 - 3x = 0$
 - e $4x^2 + 3x = 5$
 - f $-2x^2 + 9x + 5 = 0$
 - g $3x^2 - 8x + 2 = 9x - 7$
- 3 Find the solutions to the quadratic equation of the form $ax^2 + bx + c = 0$, with $a > 0$.
Express your answer in terms of a , b , and c .
- 4 To solve $x^2 - x = 2$, Emma substitutes $a = 1$ into the quadratic formula.
Find the value of c that she must substitute into the quadratic formula.
- 5 Consider the quadratic equation $9y^2 = 8y$. Suppose $b = 8$ in the quadratic formula.
 - a State the value of a .
 - b State the value of c .
- 6 Solve the following equations using the quadratic formula, leaving your answers in exact form.
 - a $x^2 + 11x + 28 = 0$
 - b $x^2 - 5x + 6 = 0$
 - c $4x^2 + 6x + 2 = 0$
 - d $4x^2 - 7x - 15 = 0$
 - e $x^2 + 5x + \frac{9}{4} = 0$
 - f $-10 + 23x + 5x^2 = 0$
 - g $x(x - 4) = 5$
 - h $-6 - 13x + 5x^2 = 0$
 - i $x^2 - 5x - 2 = 0$
 - j $4x^2 - x - 10 = 0$
 - k $-2x^2 - 15x - 4 = 0$
- 7 Solve the following equations using the quadratic formula, rounding your answers to two decimal places.
 - a $x^2 + 7x - 3 = 0$
 - b $5x^2 + 9x + 2 = 0$
 - c $1.8x^2 + 5.2x - 2.3 = 0$
 - d $3x(x + 4) = -3x + 4$



- 8 Using the quadratic formula, the solutions to a quadratic equation are given by:

$$x = \frac{-22 \pm \sqrt{484 - 108}}{18}$$

Find the values of a , b , and c .

- 9 Using the quadratic formula, the solutions to a quadratic equation are given by:

$$x = \frac{-5 \pm \sqrt{5^2 - 4 \cdot (-7) \cdot 10}}{2 \cdot (-7)}$$

- a Find the values of a , b , and c .
 - b Write down the standard form of the quadratic equation that has the given solutions.
- 10 Consider the function $y = x^2 + 8x + 12$.
- a Determine the axis of symmetry.
 - b Find the coordinates of the vertex of the parabola.
 - c Find the zeros of the quadratic function using the quadratic formula.
 - d Write down the quadratic equation in factored form.
- 11 An object is launched from a height of 90 ft with an initial velocity of 131 ft/s, and after x seconds, its height (in feet) is given by $h = -16x^2 + 131x + 90$.
Solve for the number of seconds x after which the object is 20 ft above the ground. Give your answer to the nearest tenth of a second.